

01 THE BIG PICTURE

★ Complete formula & concept map — 6-page rapid revision for JEE Main + Advanced

Reactants $\rightleftharpoons$ Products

Dynamic equilibrium — forward & backward proceed together in a closed system.

closed system

macroscopic props constant

$\Delta G = 0$

catalyst: faster, no shift



■ Law of Mass Action

for $aA + bB \rightleftharpoons cC + dD$

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$K_p = \frac{p_C^c p_D^d}{p_A^a p_B^b}$$

activities of pure solids & liquids = 1 → omitted

■ K_p - K_c Bridge

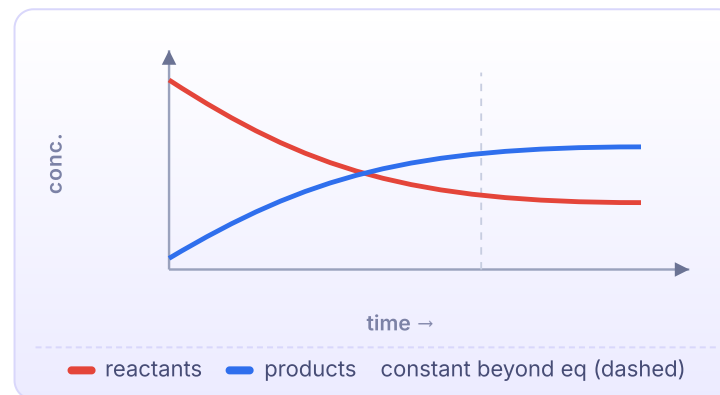
$$K_p = K_c (RT)^{\Delta n_g}$$

$$\Delta n_g = (\text{mol gas})_{\text{prod}} - (\text{mol gas})_{\text{react}}$$

$$\Delta n_g > 0 : K_p > K_c$$

$$\Delta n_g < 0 : K_p < K_c$$

$$\Delta n_g = 0 : K_p = K_c$$



02 THE K FAMILY & REACTION QUOTIENT

■ Relating K_p and K_c

$$K_p = K_c (RT)^{\Delta n_g}$$

$$\Delta n_g = (\text{gas mol})_{\text{prod}} - (\text{gas mol})_{\text{react}}$$

K depends on T only

■ Units of K

$$K_c : (\text{mol L}^{-1})^{\Delta n_g}$$

$$K_p : (\text{bar})^{\Delta n_g}$$

Thermodynamic $K = \prod a_i^{\nu_i}$ uses **activities** $\Rightarrow$ always **unitless**. $a_{\text{gas}} = \frac{p_i}{p^\circ}$, $a_{\text{soln}} = \frac{[i]}{c^\circ}$

■ Algebra of K

$$\text{reverse: } K' = \frac{1}{K}$$

$$\times n: K' = K^n$$

$$\text{halve: } K' = \sqrt{K}$$

$$\text{add steps: } K = K_1 \cdot K_2$$

■ Reaction Quotient Q

same form as K but with **any** instantaneous state. Compare $Q \leftrightarrow K$ to predict direction.

$Q < K$
net **forward** $\rightarrow$

$Q = K$
at equilibrium

$Q > K$
 $\leftarrow$ **backward**

■ Direction $\rightleftharpoons$ Spontaneity

$$\Delta G = \Delta G^\circ + RT \ln Q = RT \ln \frac{Q}{K}$$

$$Q < K \Rightarrow \Delta G < 0 \text{ (forward)}$$

$$Q > K \Rightarrow \Delta G > 0 \text{ (reverse)}$$

■ Extent from K size

$K \gg 1$: products favoured

$K \approx 1$: comparable

$K \ll 1$: reactants favoured

Independent of K : initial amounts, total P , catalyst, inert gas. **Changes K** : only temperature.

03 DEGREE OF DISSOCIATION & STANDARD SYSTEMS

Degree of Dissociation α

$$\alpha = \frac{\text{moles dissociated}}{\text{initial moles}}$$

Vapour-density / molar-mass method (n = gaseous product moles per reactant mole):

$$\alpha = \frac{D - d}{(n - 1)d}$$

$$M_{\text{mix}} = \frac{M_0}{1 + (n - 1)\alpha}$$

D =normal VD, d =observed VD, M_0 =formula mass • valid only for $1 \rightarrow n$ (one reactant $\rightarrow n$ gaseous product moles)

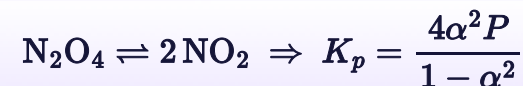
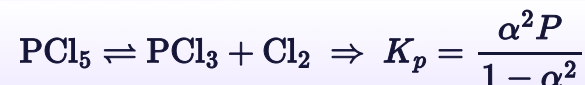
ICE — the working template

$A \rightleftharpoons nB$, start 1 mol, dissociation α , total = $1 + (n - 1)\alpha$

	A	B	Σ
Initial	1	0	1
Change	$-\alpha$	$+n\alpha$	
Equil.	$1 - \alpha$	$n\alpha$	$1 + (n - 1)\alpha$

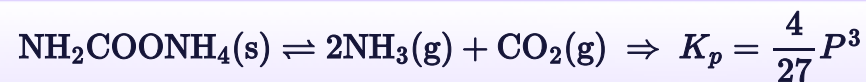
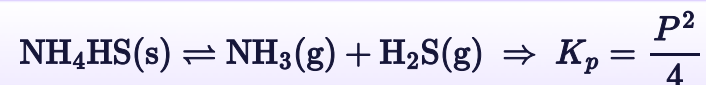
then $p_i = \frac{n_i}{\Sigma n} P \rightarrow$ build K_p .

Classic Gas Dissociations



☐ colourless $\text{N}_2\text{O}_4 \xrightleftharpoons{\text{heat}}$ brown NO_2 ☒ heat $\rightarrow$ deeper brown ($\Delta H > 0$)
if $\alpha \ll 1$: $K_p \approx \alpha^2 P$ and $4\alpha^2 P$ respectively

Heterogeneous (solid $\rightleftharpoons$ gas)



(P = total pressure from pure solid)

Simultaneous Equilibria — two solids, one common gas



common gas $X \Rightarrow$ mass balance $p_X = p_Y + p_Z$

$$p_X = \sqrt{K_{p_1} + K_{p_2}}$$

$$P_{\text{total}} = 2\sqrt{K_{p_1} + K_{p_2}}$$

e.g. NH_4HS & NH_4Cl (common NH_3)

common-gas effect: each dissociation suppressed

04 LE CHATELIER'S PRINCIPLE

Stress a system at equilibrium → it shifts to partly undo the stress.

① Concentration

shifts to consume added species

add reactant / remove product → forward

add product / remove reactant → backward

add / remove pure solid or liquid → no shift ($a = 1$)

② Pressure / Volume

to fewer gas moles

$\uparrow P$ ($\downarrow V$) → side with fewer n_{gas} $\Delta n_g = 0$ → no shift

③ Temperature (only this changes K)

endo direction on heating

$\uparrow T$ → endothermic direction $\downarrow T$ → exothermic direction

④ Catalyst

no shift

speeds up R_f & R_b equally → equilibrium reached sooner; K & position unchanged.

■ Inert Gas — the classic trap

constant V

$\rightarrow$
 $\leftarrow$
 p_i unchanged → NO shift

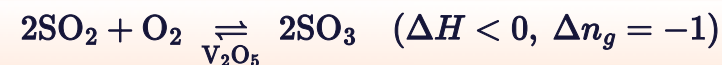
constant P

→ more n_{gas}
 $V \uparrow$ → shift to more gas moles

■ Worked read: Haber vs Contact



$\uparrow P$ helps $\downarrow T$ helps remove NH_3 cat: Fe · promoters K_2O , Al_2O_3



$\uparrow P$ helps $\downarrow T$ helps excess O_2 cat: V_2O_5

Real plants compromise T upward for rate (kinetics vs yield trade-off).

05 THERMODYNAMICS & TEMPERATURE

Free Energy $\rightleftharpoons K$

$$\Delta G^\circ = -RT \ln K = -2.303 RT \log_{10} K$$

$$\Delta G^\circ < 0 \Rightarrow K > 1$$

$$\Delta G^\circ = 0 \Rightarrow K = 1$$

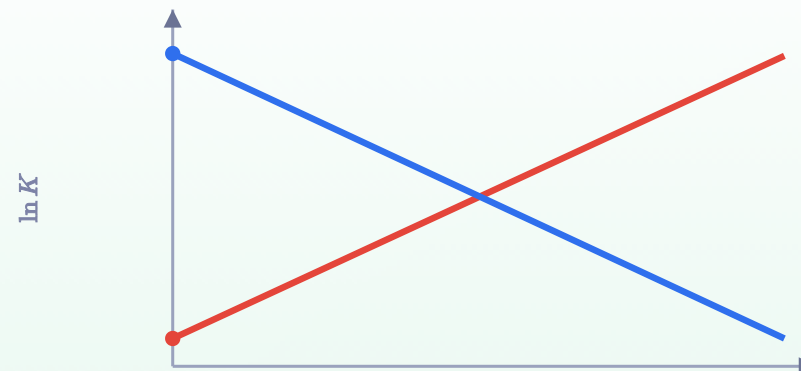
$$\Delta G^\circ > 0 \Rightarrow K < 1$$

van 't Hoff — K vs T

$$\frac{d \ln K}{dT} = \frac{\Delta H^\circ}{RT^2}$$

$$\ln K = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

$$\ln \frac{K_2}{K_1} = -\frac{\Delta H^\circ}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$



$1/T \rightarrow$

— exo ($\Delta H^\circ < 0$) : slope = $-\frac{\Delta H^\circ}{R} > 0$

— endo ($\Delta H^\circ > 0$) : slope < 0 • at $1/T \rightarrow 0$: y-intercept = $\frac{\Delta S^\circ}{R}$

Effect of heating (read from sign of ΔH°)

endothermic: $T \uparrow \Rightarrow K \uparrow$ (forward)

exothermic: $T \uparrow \Rightarrow K \downarrow$ (backward)

$\Delta H^\circ = 0$: K independent of T

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

06 RAPID REVISION & EXAM TRAPS

Must-know set

$$K_p = K_c(RT)^{\Delta n_g}$$

$$\Delta G = \Delta G^\circ + RT \ln Q$$

$$\Delta G^\circ = -RT \ln K$$

$$\alpha = \frac{D - d}{(n - 1)d}$$

Trap sheet

Drop pure solids/liquids from K ($a = 1$).

Δn_g counts **gases only**.

Inert gas @ const V : **no shift**.

Catalyst: no shift, K same.

Only T changes K .

$\alpha \ll 1$ approx only when valid.

Sign map

ΔG°	K	favoured
< 0	> 1	products
$= 0$	$= 1$	—
> 0	< 1	reactants

$K \gg 1$ fast? No — K is **thermodynamic**, not kinetic.

Le Chatelier at a glance

↑ reactant → forward

↑ product → backward

↑ P → fewer gas moles

heat + endo → forward

heat + exo → backward

inert @ const P → more gas moles

Special K_p recall

$$\text{CaCO}_3: K_p = p_{\text{CO}_2}$$

$$\text{NH}_4\text{HS}: K_p = \frac{P^2}{4}$$

$$\text{carbamate: } K_p = \frac{4}{27} P^3$$

$$R = 0.0831 \text{ L}\cdot\text{bar}\cdot\text{mol}^{-1}\text{K}^{-1}$$